Outline of the lecture

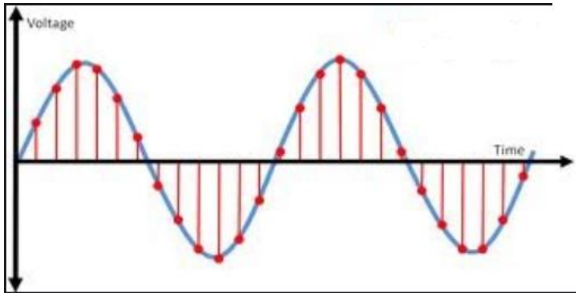
- **TYPES OF SIGNALS**
  - **COMPARISON OF DIGITAL AND ANALOG SIGNALS**
  - **DIGITAL SIGNALS**
    - ✓ Positive Logic System vs. Negative Logic System
    - ✓ Advantages
    - ✓ Disadvantages
    - ✓ Applications
- **NUMBER SYSTEMS**
  - Concept of Radix/Base
  - Various Number Systems
    - ➢ Decimal
    - ➢ Binary
    - ➢ Octal
    - ➢ Hexadecimal
  - Conversion between the different radix
- **PROBLEMS FOR PRACTICE**

Physical quantities are represented by signals.

COMPARISON

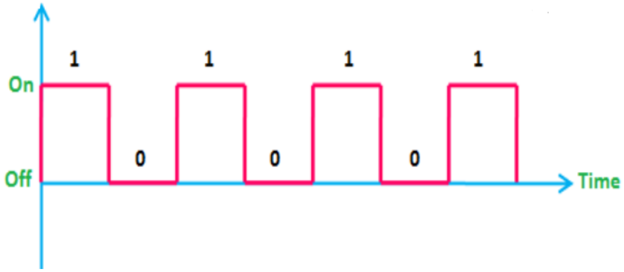
ANALOG

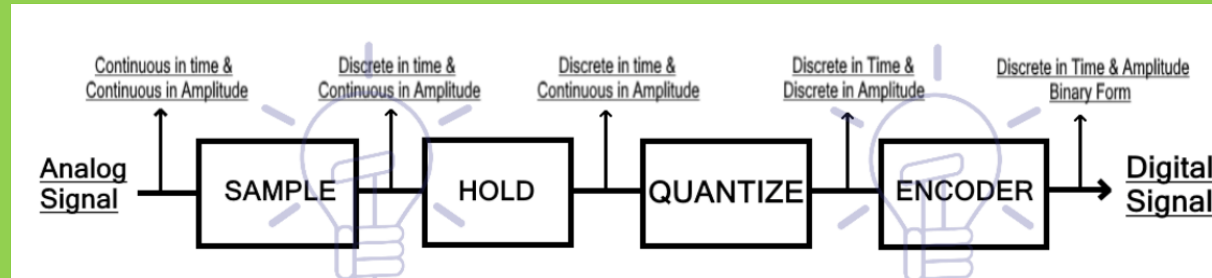
- Voltage /Current value, which is proportional to the quantity it represents (varying smoothly and continuously).
- is continuous in nature, and has an infinite set of possible values (gradually).
- Proportional in nature.
- E.g. audio signals (human voice), video signals etc.



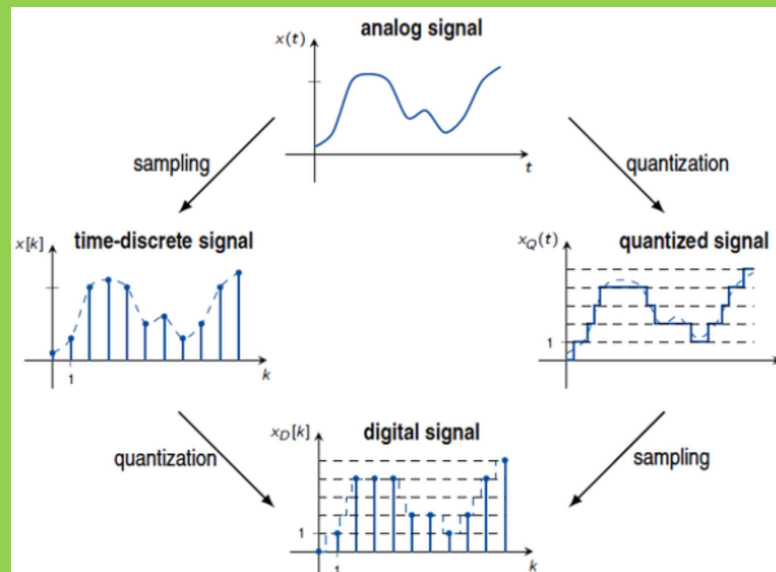
DIGITAL

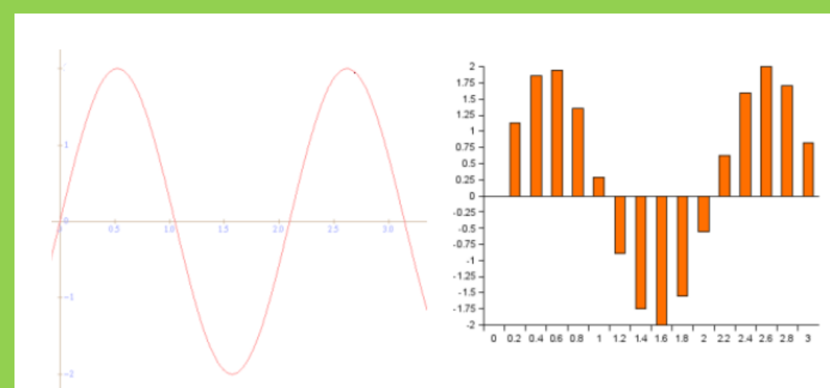
- Series of pulses of rapidly changing levels of voltage / current.
- is discontinuous in nature and rapidly changes its value within the discrete levels/steps (abruptly).
- yes/no, true/false, on/off, high/low.
- E.g. Computers, Digital Pens etc.





ANALOG TO DIGITAL CONVERSION



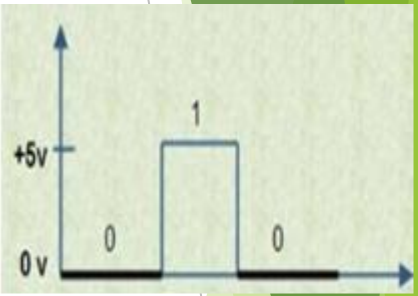


DIGITAL SIGNAL

- Comprises of discrete elements of information, i.e., by counting digits.
- In a digital system, two voltage levels represent two binary digits, 1 and 0. Based on this binary system can be classified as:

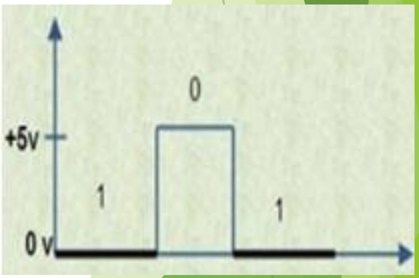
    - <u>Positive Logic System</u>: Higher voltage represented by 1

      Lower voltage represented by 0



    - <u>Negative Logic System</u>: Higher voltage represented by 0

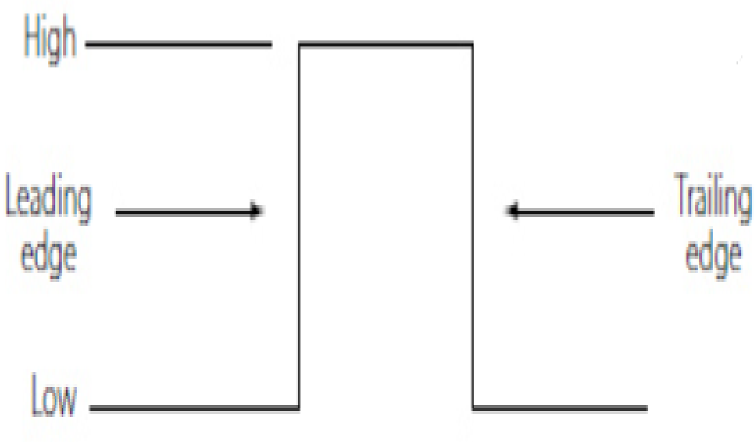
      Lower voltage represented by 1



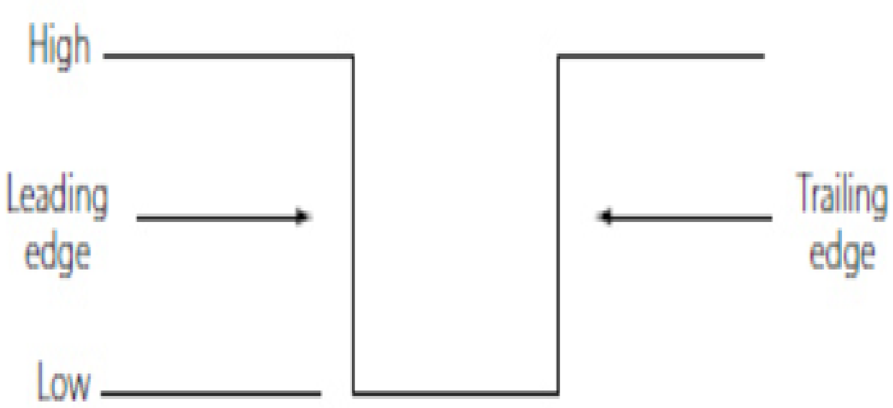
** positive logic systems are most commonly used because of ease of application.

DIGITAL SIGNALS (PULSES)

POSITIVE PULSE



NEGATIVE PULSE



A pulse has two edges:

1. Leading Edge/Rising Edge
2. Trailing Edge/Falling Edge

DIGITAL SIGNALS

ADVANTAGES

- Simple operation (only two levels)
- Few operations (ON or OFF)
- Reliable, fast and accurate
- Effect of change in temperature, ageing of components, noise is very small
- Memory capability, making it suitable for computers, calculators etc.
- Convenience (Direct display of data is convenient to read e.g. Digital Multimeter)
- Versatility

DISADVANTAGES

- Most of the natural signal are analog signals, which requisites for conversion.

DIGITAL SIGNALS

APPLICATIONS (Digital Circuits)

- Computers
- Telephony
- Data Processing
- Radar Navigation
- Military Systems
- Medical Instruments
- Consumer Products

NUMBER SYSTEMS

- Representation of any number in a number system depends on the radix/base.
- Radix/Base is the number of different digits which can occur in each position in the number system.
- Three important characteristics of a number system having positional notation (place value, i.e., each digit position has a weight or a value associated):
  - base is equal to the number of digits.
  - largest digit is one less than the base.
  - each digit is multiplied by the radix raised to an appropriate power (depending on its position) to get its place value.

Any number in any system having base r can be written as:

$$a_n a_{n-1} a_{n-2} a_{n-3} \ldots\ldots\ldots\ldots a_0 a_{-1} a_{-2} \ldots a_{-m}$$

MSB LSB

And interpreted as:

$$Y = a_n * r^n + a_{n-1} * r^{n-1} + a_{n-2} * r^{n-2} + \ldots\ldots + a_0 * r^0 + a_{-1} * r^{-1} + \ldots\ldots + a_{-m} * r^{-m}$$

Where,

Y is the value of the entire number

a_n is the value of n^{th} digit from the decimal point

NUMBER SYSTEMS TO STUDY

NUMBER SYSTEM	RADIX (BASE)	SYMBOL	EXAMPLE
Decimal	10	D	$(25)_{10}$
Binary	2	B	$(011010)_{2}$
Octal	8	O	$(256)_{8}$
Hexa-decimal	16	H	$(26FA)_{16}$

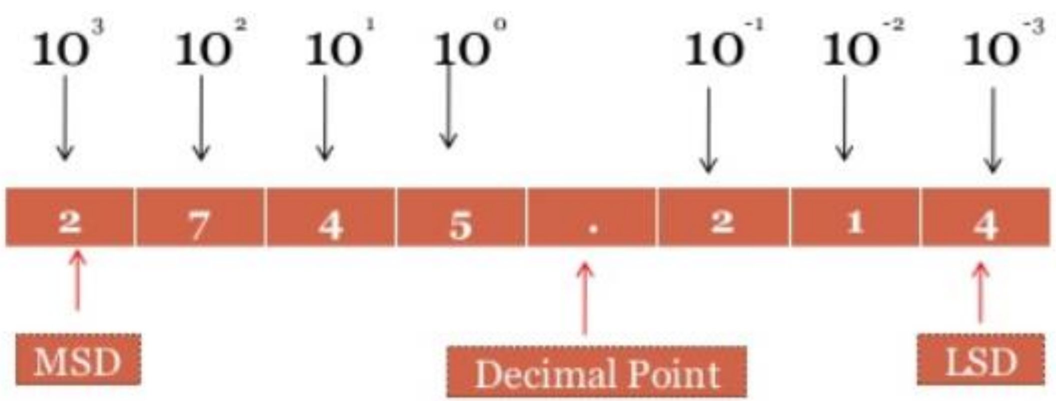
DECIMAL NUMBER SYSTEM

Base 10

- **Most commonly used.**
- **Consists of ten digits/symbols called Arabic numerals:**

  0,1,2,3,4,5,6,7,8,9(highest possible number)

- **it's a positional value system, i.e. value of a digit depends upon its position.**



$$10^{5} a_5 + 10^{4} a_4 + 10^{3} a_3 + 10^{2} a_2 + 10^{1} a_1 + 10^{0} a_0 + 10^{-1} a_{-1} + 10^{-2} a_{-2} + 10^{-3} a_{-3}$$

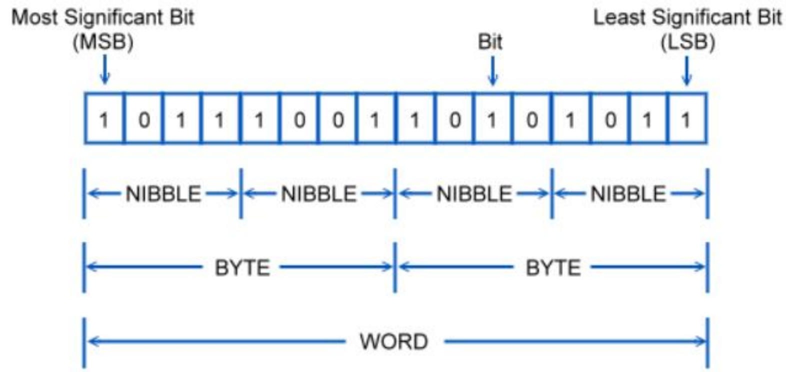
Base 2

BINARY NUMBER SYSTEM

Drawback:

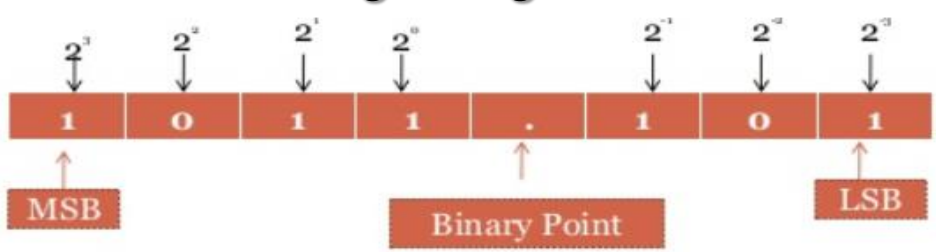
Larger numbers represented in very long sequences of 0 and 1 become very tedious.

- Most simple.
- Consists of two **binary digits (bits)**:



- It's a positional value system, i.e. value of a digit depends upon its position.
- position of a digit in binary number, indicates its weight within the number.
- Highest binary number that can be represented by n-bits binary numbers is:

$2^n - 1$ (beginning with 0)



BINARY NUMBER SYSTEM

Base 2

Decimal	MSB	4 Bit Binary		LSB
	$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

Base 2

BINARY NUMBER SYSTEM : BINARY ARITHMETIC

	Addition	Subtraction	Multiplication	Division
i)	0 + 0 = 0	0 − 0 = 0	0 x 0 = 0	0 / 1 = 0
ii)	0 + 1 = 1	1 − 0 = 1	0 x 1 = 0	1 / 1 = 1
iii)	1 + 0 = 1	1 − 1 = 0	1 x 0 = 0	0 / 0 = not allowed (not valid)
iv)	1 + 1 =10	0 - 1 = 10 − 1 (with borrow 1) = 1	1 x 1 = 1	1 / 0 = not allowed (not valid)

OCTAL NUMBER SYSTEM

Base 8

- **One of the short ways to represent binary numbers by their octal equivalents.**
- **Consists of eight digits/symbols:**

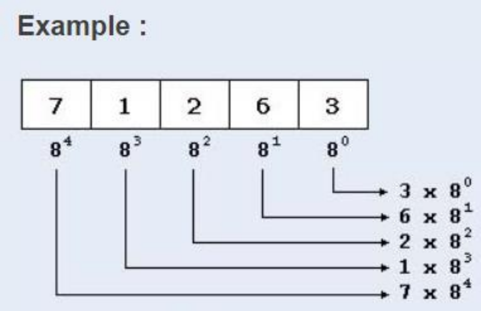
  0,1,2,3,4,5,6 and 7(highest possible number)

- **it's a positional value system, i.e. value of a digit depends upon its position.**

$$etc \ldots\ldots\ldots 8^3 \quad 8^2 \quad 8^1 \quad 8^0 \;.\; 8^{-1} \quad 8^{-2} \quad 8^{-3} \ldots\ldots\ldots etc$$

Octal Point

Decimal	Octal
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	10
9	11
10	12
11	13
12	14



Base 8

OCTAL NUMBER SYSTEM: OCTAL ADDITION

- Octal addition is performed just like decimal addition, except that if a column of two addends produces a sum greater than 7, you must subtract 8 from the result, put down that result, and carry the 1.

- Example : Add $7652^8 + 4574^8$ (carries required):

	1	1		
	7	6	5	2
+	4	5	7	4
	12 - 8 = 4	12 - 8 = 4	12 - 8 = 4	
1	4	4	4	6

Base 8

OCTAL NUMBER SYSTEM:

OCTAL SUBTRACTION

The subtraction of octal numbers follows the same rules as the subtraction of numbers in any other number system. The only variation is in the quantity of ***the borrow***.

$(253)_8 - (167)_8$

$$
\begin{array}{rccc}
 & 1 & 4+8 & 3+8 \\
 & \cancel{2} & \cancel{5} & 3 \\
- & 1 & 6 & 7 \\
\hline
 & 0 & 6 & 4 \\
\hline
\end{array}
$$

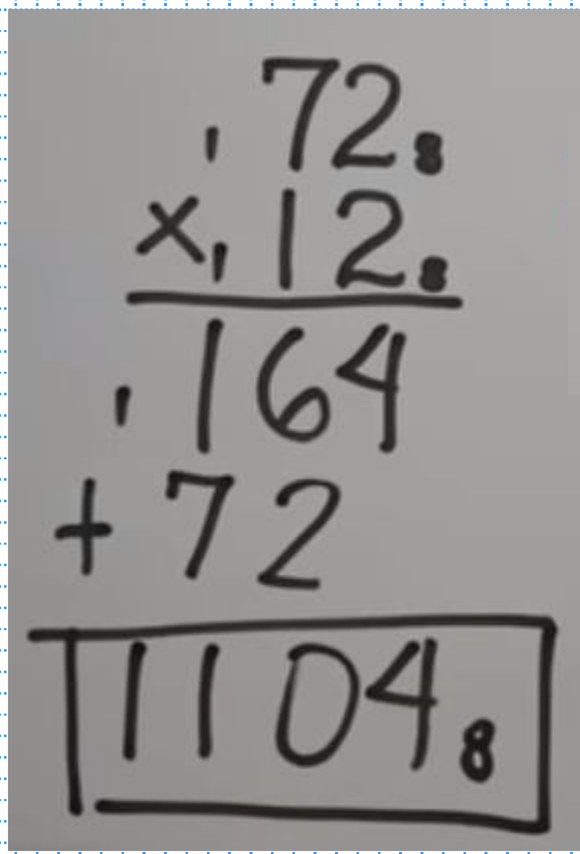
$(2304)_8 - (1065)_8$

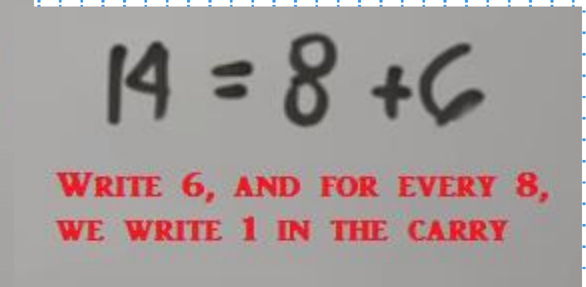
$$
\begin{array}{rcccc}
 & & 2 & \cancel{8}\ 7 & 8+ \\
 & 2 & \cancel{3} & \cancel{0} & 4 \\
- & & 0 & 6 & 5 \\
\hline
 & 2 & 2 & 1 & 7 \\
\hline
\end{array}
$$

Base 8

OCTAL NUMBER SYSTEM:

OCTAL MULTIPLICATION



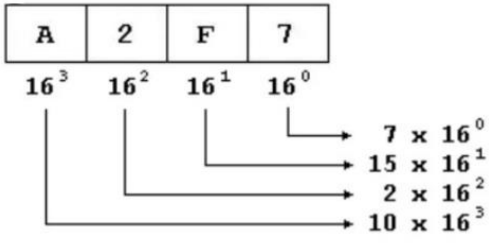


Base 8

OCTAL NUMBER SYSTEM: OCTAL DIVISION

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7
2	0	2	4	6	10	12	14	16
3	0	3	6	11	14	17	22	25
4	0	4	10	14	20	24	30	34
5	0	5	12	17	24	31	36	43
6	0	6	14	22	30	36	44	52
7	0	7	16	25	34	43	52	61

Better understood once you understand conversions.



HEXA-DECIMAL NUMBER SYSTEM

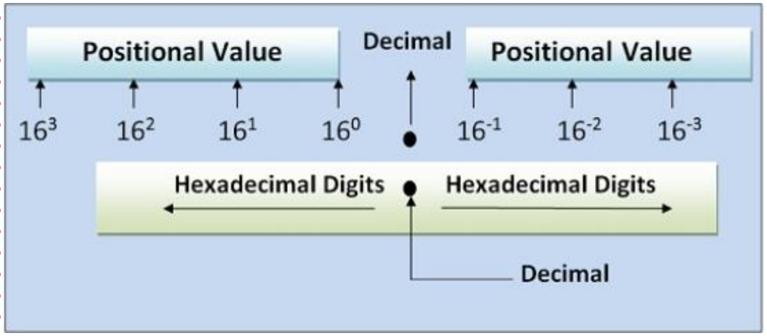
Base 16

- **Used widely in all the computer systems.**
- **Consists of 16 digits and characters:**

  0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F (highest possible number).

- **it's a positional value system, i.e. value of a digit depends upon its position.**
- **Also called Alpha-numeric number system.**

Decimal	*Hexadecimal*
0	00
1	01
2	02
3	03
4	04
5	05
6	06
7	07
8	08
9	09
10	0A
11	0B
12	0C
13	0D
14	0E
15	0F
16	10
17	11
18	12
:	:
etc.	etc.



Base 16

Hexa-Decimal Number System: Hexa-Decimal Addition

- Start with the least significant hexadecimal digits
- Let Sum = summation of two hex digits
- If Sum is greater than or equal to 16
  - Sum = Sum – 16 and Carry = 1
- Example:

```
carry         1  1     1
        9  C  3  7  2  8  6  5
  +     1  3  9  5  E  8  4  B
        ----------------------
        A  F  C  D  1  0  B  0
```

5 + B = 5 + 11 = 16
Since Sum ≥ 16
Sum = 16 – 16 = 0
Carry = 1

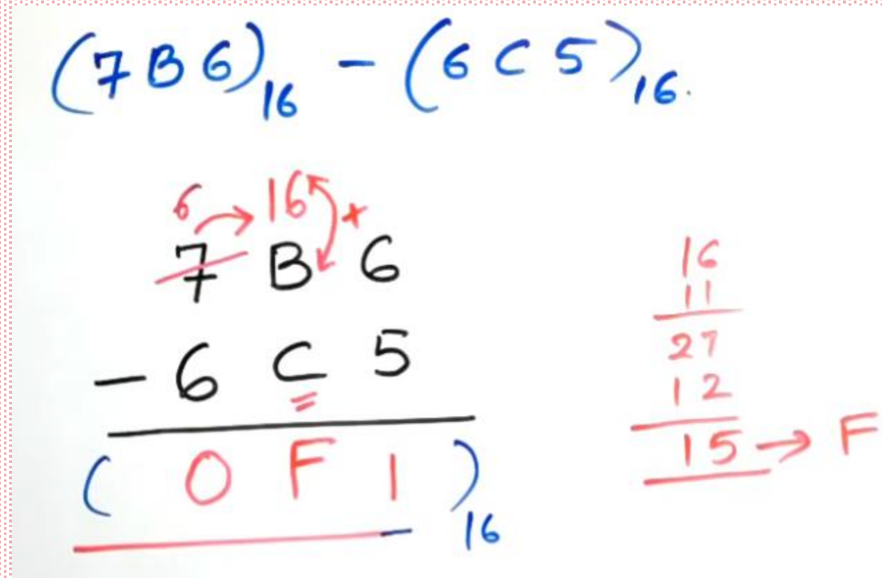
Decimal	Hexadecimal
0	00
1	01
2	02
3	03
4	04
5	05
6	06
7	07
8	08
9	09
10	0A
11	0B
12	0C
13	0D
14	0E
15	0F
16	10
17	11
18	12
:	:
etc.	etc.

```
     1   9   5
  +  3   1   9
     ---------
     4   A   E
```

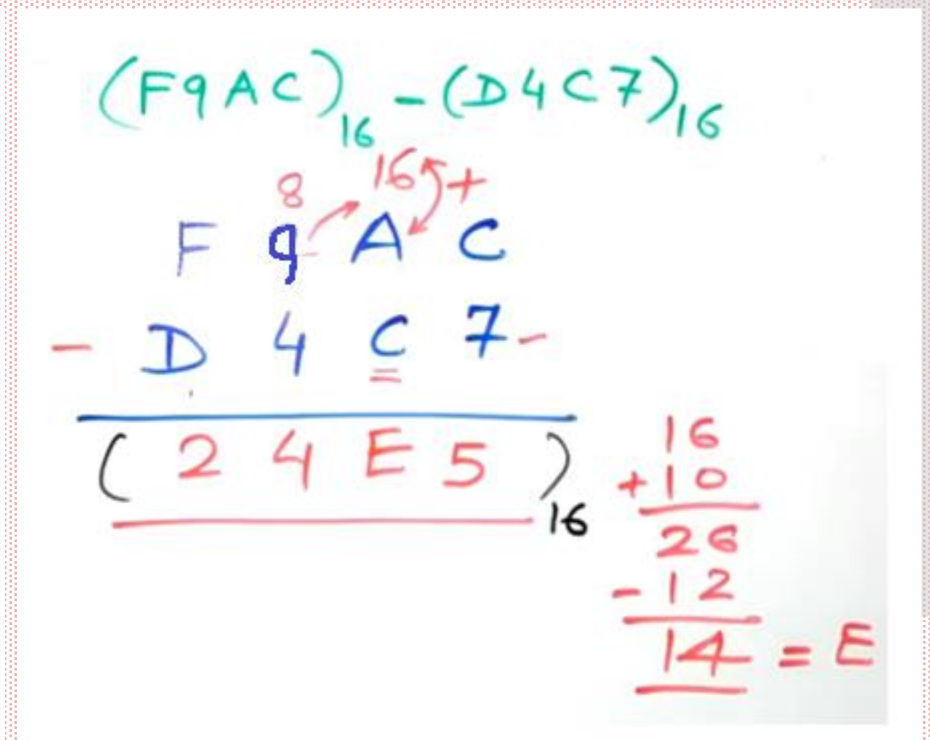
BORROW=16

Base 16

HEXA-DECIMAL NUMBER SYSTEM: HEXA-DECIMAL SUBTRACTION

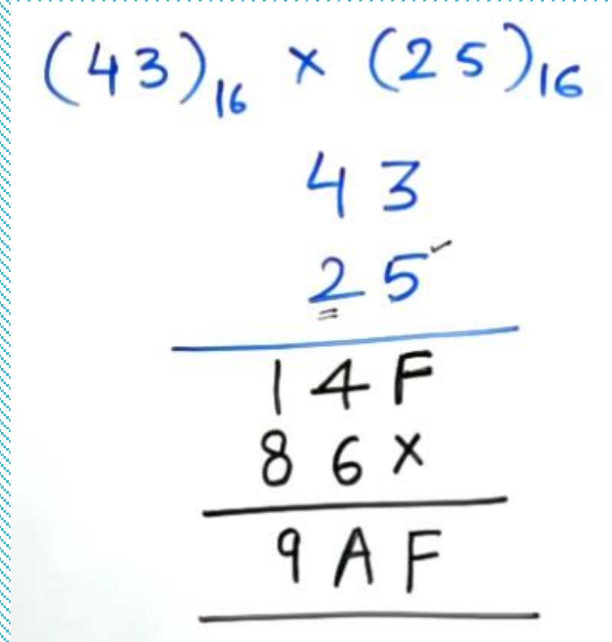


Decimal	Hexadecimal
0	00
1	01
2	02
3	03
4	04
5	05
6	06
7	07
8	08
9	09
10	0A
11	0B
12	0C
13	0D
14	0E
15	0F
16	10
17	11
18	12
:	:
etc.	etc.

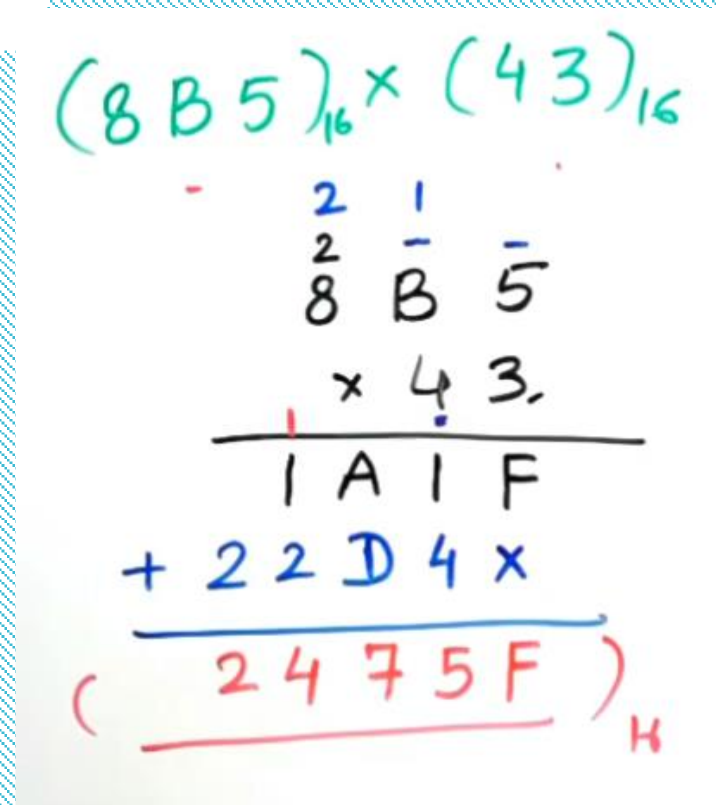


Base 16

HEXA-DECIMAL NUMBER SYSTEM: HEXA-DECIMAL MULTIPLICATION



Decimal	Hexadecimal
0	00
1	01
2	02
3	03
4	04
5	05
6	06
7	07
8	08
9	09
10	0A
11	0B
12	0C
13	0D
14	0E
15	0F
16	10
17	11
18	12
:	:
etc.	etc.



PROGRESSION AMONGST DIFFERENT NUMBER SYSTEMS

Number System	Decimal	Binary	Octal	Hexadecimal
One	1	0001	1	1
Two	2	0010	2	2
Three	3	0011	3	3
Four	4	0100	4	4
Five	5	0101	5	5
Six	6	0110	6	6
Seven	7	0111	7	7
Eight	8	1000	10	8
Nine	9	1001	11	9
Ten	10	1010	12	A
Eleven	11	1011	13	B
Twelve	12	1100	14	C
Thirteen	13	1101	15	D
Fourteen	14	1110	16	E
Fifteen	15	1111	17	F
Sixteen	16	10000	20	10

CONVERSIONS BETWEEN BASES

- Converting Any Base to Decimal
- Converting Any Base to Binary
- Converting Any Base to Octal
- Converting Any Base to Hexa-Decimal

CONVERTING ANY BASE TO DECIMAL

A number (N) can be written as:

$N = N_i + N_F$

$N_i \rightarrow$ Integer part

$N_F \rightarrow$ Fractional part

CONVERSIONS BETWEEN BASES

Converting from ANY base to decimal is done by multiplying each digit by its weight and summing.

Binary to Decimal

Octal to Decimal

Hexa-Decimal to Decimal

Convert $(350.325)_8 = (\quad)_{10}$

$N_i = 350 \quad N_F = 0.325$

$(350.325)_8 = 3\times8^2+5\times8^1+0\times8^0 \ . \ 3\times8^{-1}+2\times8^{-2}+5\times8^{-3}$

$= 192+40+0 \ . \ 0.375+0.031+0.009$

$= (232.415)_{10}$

Convert $(101111.101)_2 = (\quad)_{10}$

$N_i = 101111 \quad N_F = 0.101$

$= 1\times2^5+0\times2^4+1\times2^3+1\times2^2+1\times2^1+1\times2^0 \ . \ 1\times2^{-1}+0\times2^{-2}+1\times2^{-3}$

$= 32+0+8+4+2+1 \ . \ 0.5+0+0.125$

$= (47.625)_{10}$

Convert $(5F2C.8E)_{16} = (\quad)_{10}$

$N_i = 5F2C \quad N_F = 0.8E$

$(5F2C.8E)_{16} = 5\times16^3+F\times16^2+2\times16^1+C\times16^0 \ . \ 8\times16^{-1}+E\times16^{-2}$

$= 20480+(15\times256)+32+(12\times1) \ . \ 0.5+0.585$

$= 20480+3840+32+12 +1 \ . \ 085$

$= (24365.085)_{10}$

CONVERSIONS BETWEEN BASES

CONVERTING ANY BASE TO BINARY

- **Double Dabble Method/Repeated division by Two**
- **For N_i, keep dividing by two, keep writing the remainders until the quotient becomes 0. Take the remainders in reverse order.**
- **For N_F, multiply only fraction by 2, recording the integer part.**

Decimal to Binary

Convert $(17.8125)_{10} = (\quad)_2$

$N_i = 17$

2	17	1
2	8	0
2	4	0
2	2	0
	1	

$N_F = 0.8125$

$= 0.8125 \times 2 = 1.625$ 1

$= 0.625 \times 2 = 1.25$ 1

$= 0.25 \times 2 = 0.5$ 0

$= 0.5 \times 2 = 1.0$ 1

$(17.8125)_{10} = (10001.1101)_2$

Octal to Binary

Convert $(35.346)_8 = (\quad)_2$

$N_i = 35$ $N_F = 0.346$

Octal Point

Octal Number :	3	5	.	3	4	6
Binary Number :	011	101	.	011	100	110

Binary Point

$(35.346)_8 = (11101.011100110)_2$

<u>**Each digit is replaced by 3-bit binary equivalent**</u>

Hexa-Decimal to Binary

<u>**Each digit is replaced by 4-bit binary equivalent**</u>

Convert $(9AF.A)_{16} = (\quad)_2$

$N_i = 9AF$ $N_F = 0.A$

Hexadecimal number :	9	A	F	.	A
Binary Number :	1001	1010	1111		1010

$(9AF.A)_{16} = (100110101111.1010)_2$

CONVERSIONS BETWEEN BASES

CONVERTING ANY BASE TO OCTAL

- Double Dabble Method/Repeated division by Eight.(Decimal)
- Starting from LSB, each group of 3-bits Is replaced by its decimal equivalents. (Binary)
- Convert Hex → Binary → Octal.

Decimal to Octal

Convert $(225.225)_{10} = (\quad)_8$

$N_i = 225$ $N_F = 0.225$

$$
\begin{array}{r|l}
8 & 225 \\
8 & 28 \quad 1 \\
8 & 3 \quad 4 \\
 & 0 \quad 3
\end{array}
$$

Fractional part:

0.225 x 8 = 1.800	1
0.800 x 8 = 6.400	6
0.400 x 8 = 3.200	3
0.200 x 8 = 1.600	1
0.600 x 8 = 4.800	4

$(225.225)_{10} = (341.16314)_8$

Binary to Octal

Convert $(1101011.00101)_2 = (\quad)_8$

$N_i = 1101011$ $N_F = 0.00101$

1101011.00101

001101011.001010

$= (153.12)_8$

$(1101011.00101)_2 = (153.12)_8$

Hexa-Decimal to Octal

Convert first to binary and then Octal.

CONVERTING ANY BASE TO HEXA-DECIMAL

CONVERSIONS BETWEEN BASES

- Double Dabble Method/Repeated division by 16 (Decimal)
- Starting from LSB, each group of 4-bits Is replaced by its decimal equivalents. (Binary)
- Convert Octal → Binary → Hexa-Decimal.

Decimal to Hexa-Decimal

Binary to Hexa-Decimal

Each digit is replaced by 4-bit decimal equivalent

Octal to Hexa-Decimal

Convert first to binary and then Hexa-Decimal.

Convert $(374.37)_{10} = (\quad)_{16}$

$N_i = 374$ $\quad$ $N_F = 0.37$

16	374	
16	23	6
16	1	7
	0	1

$(176)_{16}$

$0.37\times16 = 5.92 = 0.92$ with Carry 5

$0.92\times16 = 14.72 = 0.72$ with Carry 14 (E)

$0.72\times16 = 11.52 = 0.52$ with Carry 11 (B)

$0.52\times16 = 8.32 = 0.32$ with Carry 8

$(0.5EB8)_{16}$

$(374.37)_{10} = (176.5EB8)_{16}$

Convert $(1101011.0010)_2 = (\quad)_{16}$

$N_i = 1101011$ $\quad$ $N_F = 0.0010$

1101011.0010

01101011.0010

$= 0\times2^3+1\times2^2+1\times2^1+0\times2^0 \;\; 1\times2^3+0\times2^2+1\times2^1+1\times2^0 \;.\; 0\times2^3+0\times2^2+1\times2^1+0\times2^0$

$= 6B.2$

$(1101011.0010)_2 = (6B.2)_{16}$

QUESTION BANK

Decimal to Others

i). $(549.58)_{10} = (\ ?\ )_2$

ii). $(20.25)_{10} = (\ ?\ )_2$

iii). $(6745.85)_{10} = (\ ?\ )_8$

iv). $(5896.72)_{10} = (\ ?\ )_{16}$

Binary to Others

v). $(10111001.101)_2 = (\ ?\ )_{10}$

vi). $(1101110.0101)_2 = (\ ?\ )_8$

vii). $(111010110.10111)_2 = (\ ?\ )_{16}$

viii). $(1100110110.1111)_2 = (\ ?\ )_{16}$

Octal to Others

ix). $(47365.56)_8 = (\ ?\ )_{10}$

x). $(152753.41)_8 = (\ ?\ )_2$

xi). $(356211.25)_8 = (\ ?\ )_{16}$

xii). $(551267.75)_8 = (\ ?\ )_{16}$

Hexadecimal to Others

xiii). $(47A3F5.D6)_{16} = (\ ?\ )_{10}$

xiv). $(1BD29E.6C)_{16} = (\ ?\ )_2$

xv). $(7FA554.80D)_{16} = (\ ?\ )_8$

xvi). $(9C1A00.0E)_{16} = (\ ?\ )_8$

THANK YOU

Base 16

Hexa-decimal Number System: Hexa-decimal Division

```
            1   0   B
2  B | 2   C   F   3
       2   B
       ---------
           1   F   3
           1   D   9
           -------------
Remainder      1   A
```

```
                         F  .F   F
1  0  0  0 | F   F   F   F  .0   0
             F   0   0   0
             ---------------
                 F   F   F   0
                 F   0   0   0
                 ---------------
                     F   F   0   0
                     F   0   0   0
                     ---------------
            Remainder    F   0   0
```

Decimal	Hexadecimal
0	00
1	01
2	02
3	03
4	04
5	05
6	06
7	07
8	08
9	09
10	0A
11	0B
12	0C
13	0D
14	0E
15	0F
16	10
17	11
18	12
:	:
etc.	etc.